

Arjun Patel

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EDUCATION

University of California, Berkeley | GPA: 3.92/4.00 | Berkeley, CA Dec. 2026
Double Major: Astrophysics & Applied Mathematics Concentrating in Quantitative Analysis
Brighton High School | GPA: 3.98/4.00 | ACT: 36 | Rochester, NY Jun. 2023

PROFESSIONAL EXPERIENCE

Quantitative Trading Intern *Flow Traders Ascent* | New York, NY Apr. 2025 - Present

- Collaborated with peers in simulated trading competition and received mentorship from trading professionals.
- Trained on market-making strategies, high-frequency trading, and real-time risk management.

Research Assistant *Rubin Observatory* | Berkeley, CA Aug. 2024 - Present

- Engineered a pipeline to analyze and compare telescope chamber temperatures across different date ranges.
- Created visualizations with heat-map overlays to showcase atmospheric distortions in star imaging over time.
- Conducted Fourier transforms on thermal images to detect the frequency spectrum of stars over time.
- Collaborating with Berkeley Dean Steven Kahn and Stanford PhD candidate on data analysis.

Fellow *NASA Ames Research Center* | Berkeley, CA Sep. 2024 - May. 2025

- Assessing technology gaps & ISAM applications in small satellite advancements for NASA missions.
- Analyzing survey data from commercial, academic, and government sectors & interviewing aerospace engineers.
- Formulating strategies to monitor and resolve key technology gaps in small satellite development.

Data Analyst *NASA SPORES - HWO* | Berkeley, CA Jan. 2025 - May. 2025

- Processed radial velocity data with Python to refine exoplanet mass and orbit constraints.
- Identified and standardized datasets from major spectrographs like HIRES/Keck and HARPS/ESO.
- Published in an upcoming research paper for contributions to exoplanet detection research.

Data Science & Engineering Intern *Dotlas* | Berkeley, CA Oct. 2023 - Jan. 2025

- Developed web scrapers in Python for ToastTab, Booking.com, HERE traffic, The Entertainer, and Mcdonalds.
- Designed official data visualization website using Streamlit to enhance internal data accessibility.
- Built a pipeline to automatically extract social media URLs from any site, streamlining data collection.
- Created LLM - based prompts and agents to return organized promotional content and event details.
- Architected a competitor indexing system to identify and rank leading competitors across various brands.

Research Assistant *University of Rochester LLE* | Rochester, NY Jun. 2022 - Sep. 2022

- Modeled experiments using COMSOL Multiphysics to detect electromagnetic radiation in biomedical applications.
- Investigated terahertz radiation for differentiating between healthy and cancerous tissues with high precision.

EXTRACURRICULAR ACTIVITIES

Telescope Officer *Undergraduate Astronomy Society* | Berkeley, CA Aug. 2023 - Present

- Trained new members in operating a variety of telescopes, including Treffers, Cassegrain, Newtonian, and EV.
- Managed telescope operations for ten events per semester, ensuring precise alignment and optimal performance.

Competitor *Poker @ Berkeley* | Berkeley, CA Sep. 2023 - Present

- Placed 3rd in the Jane Street Estimathon Contest hosted at Berkeley by solving real-world estimation problems.
- Compete in weekly poker sessions, club-hosted events, poker class, and several tournaments across California.

PROJECTS

Adaptive ML - Based Momentum & Mean Reversion Trading

- Trained machine learning model to detect trends & switch between momentum and mean-reversion strategies.
- Engineered features like rolling volatility, z-scores, and stop loss risk to optimize signal accuracy.
- Outperformed the S&P 500 by 5× over the past five years through dynamic strategy selection.

COURSEWORK, SKILLS & INTERESTS

Coursework: Quantum Mechanics, Probability & Random Processes, Real Analysis, Abstract Algebra, Stellar Physics, Combinatorics, Discrete Math, Linear Algebra & Differential Equations, Numerical Analysis, Quant Finance, Poker

Skills: Python, SQL, Matlab, Java, GitHub, Jupyter, Databricks, AWS, GCP, Git, Docker, LLMs, Microsoft Office

Interests: Astronomy, Poker, Math, Trading, VC, Lifting, Gaming, Football, Marvel, Legos, Board Games, Cooking